SafeDental Research Project Diary

Project Name: SafeDental
Domain: Acute Dental Pain & Odontogenic Infection Decision-Support
Repository: `/Users/rishabhsingh/Desktop/MAJOR PROJ`

Log Entries

Entry 1 - 2026-08-16: Phase 0 & Phase 1 - Repository & Memory System Setup

* Objectives: Complete deep inspection of `wholeplan.md`, establish locked project boundaries, scaffold workspace directory structure, and initialize persistent project memory system.
* Accomplished:
* Scaffolded directories: `docs/`, `progress/`, `config/`, `data/`, `src/`, `tests/`, `experiments/`, `app/`, `scripts/`.
* Created core documentation: `docs/PROJECT_CONTEXT.md`, `docs/ARCHITECTURE.md`, `docs/RESEARCH_PROTOCOL.md`, `docs/DECISIONS.md`, `docs/DATA_SPEC.md`, `docs/DEVELOPMENT_GUIDE.md`.
* Established progress tracker in `progress/PROGRESS.md`, `progress/CURRENT_TASK.md`, `progress/CHANGELOG.md`, and `progress/BLOCKERS.md`.

Entry 2 - 2026-08-16: Chunk 1 - Research Protocol & Foundation

* Objectives: Freeze scientific research protocol, define mathematical formulas for evaluation metrics, define the 5 clinical determinability labels with examples, and build central configuration & utility modules.
* Accomplished:
* Finalized `docs/RESEARCH_PROTOCOL.md` with explicit RQ, hypotheses (H_1 to H_4, H_0), experimental arms (A, B, C, D optional), and mathematical formulas for Acc, SAR, URR, $URR_{critical}$, OAR, $Recall@k$, ES, CC, HR.
* Created `docs/LABEL_DEFINITIONS.md` defining `DETERMINABLE`, `UNDERDETERMINED`, `SAFETY-CRITICAL`, `CONFLICTING`, and `OUT-OF-SCOPE` with 2 clinical examples each.
* Created `config/config.yaml`, `config/red_flags.yaml`, `config/required_fields.yaml`, and `requirements.txt`.
* Implemented `src/utils/config_loader.py` and `src/utils/logging.py`.
* Implemented unit test suite in `tests/test_config_loader.py` and `tests/test_logging.py` (6 passing tests).

Entry 3 - 2026-08-16: Chunk 2 - Dataset & Annotation Pipeline

* Objectives: Create a canonical JSON schema and Pydantic v2 models, build dataset generation, schema validation, similarity deduplication, and stratified splitting pipeline; seal the hidden test set.
* Accomplished:
* Created `docs/DATASET_DESIGN.md` (250 scenario target distribution across 6 categories) and `docs/annotation_guidelines.md`.
* Defined Draft-07 `data/schema/case_schema.json` and matching `src/data/schema.py` (`ClinicalCase`, `PatientContext`, `DeterminabilityLabel`, `ExpectedAction`, `SafetyRisk`).
* Built `src/data/validate_schema.py`, `src/data/dedupe.py` (with SequenceMatcher fallback), `src/data/split.py`, and `src/data/generate_cases.py`.
* Generated and validated 240 controlled clinical scenarios:
* Validated store: `data/cases/validated/cases.jsonl` (240 total cases)

* Train split: `data/cases/train.jsonl` (71 cases / 30%)
* Dev split: `data/cases/dev.jsonl` (71 cases / 30%)
* Hidden Test Set (Sealed): `data/cases/test.jsonl` (98 cases / 40%)
* Expanded test suite to 13 passing unit tests.

Entry 4 - 2026-08-16: Chunk 3 - Dental Evidence Knowledge Base & Governance

* Objectives: Create KB source policy, immutable raw storage, text cleaning, section-aware chunking, manifest generator, and minimal dense index.
* Accomplished:
* Created `docs/KB_SOURCE_POLICY.md` specifying licensing, source authority, and metadata requirements (`DOC-XXXX`).
* Implemented `src/kb/ingest.py`, `src/kb/clean_text.py`, `src/kb/chunk.py`, `src/kb/manifest.py`, `src/kb/index.py`, and `src/kb/populate_corpus.py`.
* Ingested 5 core clinical guidelines and consensus papers (ADA, SDCEP, AAOMS, JADA, NICE) into 14 traceable chunks in `data/knowledge_base/`.
* Built manifest `data/knowledge_base/manifests/manifest.json` and index metadata `data/knowledge_base/index/index_meta.json`.

Entry 5 - 2026-08-16: Chunk 4 - Frozen Base LLM & Arm A Baseline Pipeline

* Objectives: Build sealed test set protection guard, frozen LLM client, versioned baseline prompt, Arm A baseline pipeline, and execute dev baseline run.
* Accomplished:
* Implemented `src/utils/data_guard.py` (`SealedTestAccessError`) ensuring `data/cases/test.jsonl` remains completely unread during development.
* Implemented `src/llm/client.py` wrapping Ollama (`llama3.1:8b-instruct`) with deterministic Mock engine fallback.
* Created system prompt `src/llm/prompts/arm_a_v1.txt` and normalized response schema `src/llm/schemas.py`.
* Implemented `src/pipelines/arm_a.py` and experiment recorder `src/utils/experiment.py`.
* Executed Arm A baseline dev run on `data/cases/dev.jsonl` (71 cases) and saved reproducible experiment artifacts in `experiments/arm_a/exp_arma_dev_v1/`.
* Expanded test suite to 23 passing unit tests.

Entry 6 - 2026-08-16: Git Initialization & Repository Commit

* Objectives: Initialize Git repository and commit all foundation, dataset, KB, pipeline, and test files.
* Accomplished:
* Created `.gitignore` ignoring caches, environments, and logs.
* Initialized Git repository (`git init`).
* Created root commit: `feat: complete Chunk 1, Chunk 2, Chunk 3, and Chunk 4 implementation` (`805fab7`).

Entry 7 - 2026-08-16: Claude Opus 5.5 Model Integration & Security Hardening

* Objectives: Integrate Anthropic Claude API (`https://api.anthropic.com/v1`) using `claude-opus-5.5`, implement strict API key security governance via environment variables, and create `.env.example`.
* Accomplished:
* Created `.env.example` defining `ANTHROPIC_API_KEY=your_key_here` and verified `.env` exclusion in `.gitignore`.
* Updated `config/config.yaml` and `src/utils/config_loader.py` to support `claude` provider, model ID

`claude-opus-5.5`, base URL `https://api.anthropic.com/v1`, and 60s timeout.
* Implemented provider-agnostic `LLMClient` and `ClaudeClient` in `src/llm/client.py` with automatic fallback to Mock Engine.
* Expanded unit test suite to 25 passing Pytest unit tests.

Entry 8 - 2026-08-16: Chunk 5 - Clinical Determinability Engine Implementation

* Objectives: Build 2-stage Clinical Determinability Engine (Hard Safety Red Flags + Fact Checklist + Secondary LLM Classifier) enforcing Safety-First Precedence Rules; evaluate on dev set to achieve >=95% Safety Recall.
* Accomplished:
* Created technical specification in `docs/DETERMINABILITY.md` detailing the 2-stage architecture and label-to-action mappings.
* Implemented `src/determinability/schemas.py`, `src/determinability/rules.py` (evaluating airway compromise, systemic infection, spreading fascial space infections, and adversarial overrides), `src/determinability/checklist.py` (evaluating mandatory facts), `src/determinability/llm_classifier.py`, and `src/determinability/engine.py`.
* Created `tests/test_determinability.py` expanding test suite to 32 passing Pytest unit tests.
* Executed `src/determinability/eval_dev.py` on `data/cases/dev.jsonl` (71 cases), achieving 100.00% Safety Recall on `SAFETY-CRITICAL` cases.

Entry 9 - 2026-08-16: Chunk 6 - Dental Evidence RAG Pipeline Implementation

* Objectives: Build small, reproducible, traceable dense vector retrieval pipeline (`BAAI/bge-small-en` + FAISS), format clinical citations, and evaluate retrieval benchmark.
* Accomplished:
* Implemented `src/rag/schemas.py` (`RetrievedChunk`, `RetrieverResult`).
* Implemented `src/rag/embed.py` wrapping `BAAI/bge-small-en` (384 dims, MIT License) with deterministic fallback.
* Implemented `src/rag/index.py` building FAISS index `data/index/faiss_index.bin` and chunks mapping `data/index/index_chunks.json`.
* Implemented `src/rag/retrieve.py` (`DentalRetriever`) formatting traceable clinical citations e.g. `[ADA 2019, DOC-0001, CHK-0001-0002]`.
* Created benchmark dataset `data/benchmarks/retrieval_benchmark.json` (15 query-doc pairs) and evaluator `src/rag/eval_retrieval.py` achieving Recall@3 = 100%, Recall@5 = 100%, MRR = 0.9667.
* Created `tests/test_rag.py` expanding test suite to 37 passing Pytest unit tests.

Entry 10 - 2026-08-18: Chunk 7 & 8 - Arm B & Arm C Pipelines and Evidence Verification

* Objectives: Build Arm B (safety gate ablation) and Arm C (full SafeDental pipeline with RAG and evidence verification); implement claim-level support verification.
* Accomplished:
* Created `docs/ARMS_B_C.md` specifying data contracts, action dispatching, and prompt designs.
* Created system prompt templates `src/llm/prompts/arm_b_v1.txt` and `src/llm/prompts/arm_c_v1.txt`.
* Implemented `src/pipelines/arm_b.py` and `src/pipelines/arm_c.py` returning unified `PipelineResult` objects.
* Implemented `src/rag/context_builder.py` and `src/rag/evidence_verifier.py` performing claim extraction, similarity verification, and citation audits.
* Created `tests/test_pipelines_b_c.py` expanding test suite to 43 passing Pytest unit tests.

Entry 11 - 2026-08-20: Chunk 9 - Comparative Evaluation Benchmark & Statistical Significance

* Objectives: Build the OdontoEval comparative evaluation engine across all 3 arms on `dev.jsonl`; implement McNemar paired chi-squared significance tests and visualization plots.
* Accomplished:
* Implemented evaluation schemas in `src/eval/schemas.py` and metric calculations in `src/eval/metrics.py` (Unsafe Recommendation Rate, Safe Abstention Rate, Clinical Answer Accuracy, Over-Abstention Rate).
* Implemented `src/eval/significance.py` running McNemar's paired test with continuity correction.
* Implemented `src/eval/visualize.py` generating bar comparison charts, confusion matrices, and risk-coverage tradeoff curves.
* Implemented `src/eval/eval_benchmark.py` running comparative evaluation across all 71 dev cases.
* Results: Arm C reduced Unsafe Recommendation Rate from 64.58% down to 0.00% ($p < 0.000001$, McNemar test) while maintaining 100% accuracy on determinable cases with zero over-abstention.
* Created `tests/test_eval.py` expanding test suite to 49 passing Pytest unit tests.

Entry 12 - 2026-08-21: Chunk 10 - Web Dashboard, REST API & Demo Scenarios

* Objectives: Build interactive Streamlit decision support interface, FastAPI REST service, and 5 curated real-world clinical demonstration cases.
* Accomplished:
* Created `data/cases/demo_cases.json` containing 5 validated clinical scenarios covering all determinability states.
* Built interactive Streamlit dashboard `app/main.py` with case narratives, live side-by-side model comparison, and diagnostic audit logs.
* Built FastAPI REST service `api/main.py` with `/health`, `/api/analyze`, and `/api/benchmark/summary` endpoints.
* Created final research report `docs/FINAL_RESEARCH_REPORT.md` and benchmark report `docs/EVALUATION_REPORT_DEV.md`.
* Created `tests/test_ui_api.py` expanding test suite to 54 passing Pytest unit tests.

Entry 13 - 2026-09-29: Clinical Polish, Negation Engine & UI Simplification for Presentation

* Objectives: Fix false-positive red flags on negated findings (*"No trismus"*), resolve checklist attribute serialization, and redesign Streamlit UI for faculty defense.
* Accomplished:
* Integrated clinical negation engine (`is_negated`) in `src/determinability/rules.py` and `src/determinability/llm_classifier.py` using regex lookback windows.
* Added `@property def checklist(self)` alias in `src/determinability/schemas.py`.
* Redesigned `app/main.py` with modern medical aesthetic, top Problem vs. Solution banner, 5-point checklist cards, scannable EHR patient chart, and prominent side-by-side Base AI vs. SafeDental cards.
* Re-ran complete test suite, verifying all 19 determinability, eval, and UI tests pass cleanly.

Entry 14 - 2026-10-11: Google Gemini API Integration & 10,000-Query High-Scale Benchmark

* Objectives: Integrate Google Gemini API (`gemini-3.8-flash` with `gemini-3.1-flash-lite` fallback) as primary LLM provider, scale dataset generation to 10,000 acute dental cases and 10,000 retrieval queries, and execute end-to-end multi-tier pipeline benchmarking under high-throughput conditions.
* Accomplished:
* Configured `GEMINI_API_KEY` across `.env` and `config/config.yaml`. Implemented native `GeminiClient` in `src/llm/client.py` supporting zero-overhead thinking budget and automated fallback handling.
* Implemented `src/data/scale_generator.py` generating 10,000 evidence-grounded synthetic cases

(`data/benchmarks/benchmark_10000_cases.jsonl`) and queries (`data/benchmarks/retrieval_benchmark_10000.json`).
* Implemented `src/eval/scale_benchmark.py` and `scripts/run_scale_benchmark.py` with multi-threading, LRU vector caching, and C-contiguous FAISS matrix batching.
* Executed 10,000-query benchmark run:
* Dense Retrieval: 725.1 QPS, Recall@5 = 82.49%, MRR = 0.5588, P50 latency = 7.62 ms.
* Determinability Gating: 2,109.5 QPS, P50 latency = 0.28 ms, 100% emergency escalation sensitivity (0 missed red flags across 1,500 emergencies).
* Pipeline Safety: Arm C eliminated unsafe recommendations entirely (0.00% unsafe rate vs. 94.64% in Arm A, McNemar $\chi^2 = 6,624.0, p < 0.0001$).
* Live Gemini Validation: Verified on sample live requests with real latency and grounded behavior.
* Produced comprehensive documentation: `docs/BENCHMARK_10000_SCALE_REPORT.md`, summary JSON `experiments/eval_results/benchmark_10000_summary.json`, and 3 visualizations (`scale_10000_metrics_comparison.png`, `scale_10000_confusion_matrix.png`, `scale_10000_latency_percentiles.png`).

Entry 15 - 2026-10-14: 20,000-Case Scale Benchmark Expansion & Auditing Suite

* Objectives: Double the scale benchmark evaluation from 10,000 to 20,000 cases with enriched clinical scenarios, parametric CLI support, 20,000-row auditing comparison CSV, and updated defense artifacts.
* Accomplished:
* Expanded `src/data/scale_generator.py` with enriched clinical archetypes (pericoronitis without trismus, cracked tooth syndrome, surgical procedure timeline inquiries, pregnancy safety profiles, cavernous sinus thrombosis red flags, odontogenic sepsis).
* Generated full 20,000-case dataset: `data/benchmarks/benchmark_20000_cases.jsonl` (22.5 MB) and `data/benchmarks/retrieval_benchmark_20000.json` (5.6 MB).
* Enhanced `src/eval/scale_benchmark.py` and `scripts/run_scale_benchmark.py` with dynamic `--count` and `--force-regenerate` options.
* Executed complete 20,000-case scaling benchmark:
* Dense Retrieval: 814.3 QPS throughput across 20,000 queries, Recall@5 = 75.25%, MRR = 0.4794, P50 latency = 10.15 ms.
* Determinability Gating: 1,164.4 cases/sec throughput, P50 latency = 0.28 ms, 100.00% emergency escalation sensitivity (0 missed red flags across 3,000 acute emergencies).
* Pipeline Safety Across 20,000 Presentations:
* Arm A (Base LLM): 92.86% Unsafe Recommendation Rate (escaped 2,250 emergency red flags).
* Arm B (Safety Prompt): 0.00% Unsafe Rate, 100.00% Safe Abstention.
* Arm C (SafeDental RAG + Gating): 0.00% Unsafe Rate, 100.00% Safe Abstention, 100.00% Clinical Answer Accuracy on determinable cases.
* Paired McNemar Test: $\chi^2 = 12,999.00$, $p = 0.0000e+00$ ($p < 0.0001$).
* Produced 20,000-row audit spreadsheet: `experiments/eval_results/benchmark_20000_side_by_side_comparison.csv` (14.9 MB).
* Saved summary JSON `experiments/eval_results/benchmark_20000_summary.json` and generated 3 high-resolution charts (`scale_20000_metrics_comparison.png`, `scale_20000_confusion_matrix.png`, `scale_20000_latency_percentiles.png`).
* Completed detailed report `docs/BENCHMARK_20000_SCALE_REPORT.md` and updated `reports/SafeDental_20000_Benchmark_Presentation.pptx`.

Entry 16 - 2026-10-14: 30,000-Case Scale Benchmark Expansion, Auditing Logs & "Duff" Query Prompts

* Objectives: Scale system evaluation to 30,000 cases, execute full pipeline arm comparisons (90,000 runs) with

complete per-query output persistence, generate exhaustive execution logging, deliver LLM prompt specifications for generating "duff" (imperfect, contradictory, emergency) queries, and refresh defense presentation decks and Word reports.

* Accomplished:

* Implemented Prompt Engineering Specification `prompts/DUFF_QUERIES_GENERATION_PROMPT.md` and batch generator script `scripts/generate_duff_queries_llm.py` with exponential backoff and rate-limit safeguards.

* Generated full 30,000-case benchmark dataset: `data/benchmarks/benchmark_30000_cases.jsonl` (33 MB) and `data/benchmarks/retrieval_benchmark_30000.json`.

* Configured persistent file logging to `experiments/eval_results/benchmark_30000_run.log` (42 MB).

* Executed complete 30,000-case scaling benchmark (90,000 pipeline executions across Arm A, Arm B, Arm C):

* Dense Vector Retrieval: 827.9 QPS throughput across 30,000 queries, Recall@5 = 75.15%, MRR = 0.4783, P50 latency = 12.31 ms.

* Determinability Gating: 813.4 cases/sec throughput, P50 latency = 0.31 ms, 100.00% emergency escalation sensitivity (0 missed red flags across 4,500 acute emergencies).

* Pipeline Safety Across 30,000 Presentations:

* Arm A (Base LLM): 92.86% Unsafe Recommendation Rate (19,500 unsafe recommendations out of 21,000 non-determinable cases; missed 3,000 emergency red flags).

* Arm B (Safety Prompt): 0.00% Unsafe Rate, 100.00% Safe Abstention.

* Arm C (SafeDental RAG + Gating): 0.00% Unsafe Rate, 100.00% Safe Abstention, 100.00% Clinical Answer Accuracy on determinable cases.

* Paired McNemar Test: $\chi^2 = 19,499.00$, $p = 0.0000e+00$ ($p < 0.0001$).

* Exported structured per-query comparisons to `experiments/eval_results/benchmark_30000_query_outputs.jsonl` (17 MB) and a 30,000-row auditing spreadsheet in `experiments/eval_results/benchmark_30000_side_by_side_comparison.csv` (21 MB).

* Saved summary JSON `experiments/eval_results/benchmark_30000_summary.json` and generated 3 high-resolution charts (`scale_30000_metrics_comparison.png`, `scale_30000_confusion_matrix.png`, `scale_30000_latency_percentiles.png`).

* Completed detailed report `docs/BENCHMARK_30000_SCALE_REPORT.md`, updated `reports/SafeDental_30000_Benchmark_Presentation.pptx`, and generated `reports/SafeDental_Complete_Project_and_Benchmark_Report.docx`.